AI-based Student Performance Prediction and Early Warning System

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ABSTRACT – Student performance and dropout risk are influenced by a combination of academic, behavioural, psychological, and social factors that cannot be accurately assessed through conventional evaluation methods alone. This study proposes an explainable AI-based early warning framework for continuously monitoring student progress, predicting academic risk, and supporting timely educational interventions through intelligent learning analytics. The framework utilizes multi-dimensional student data, including attendance records, academic performance, homework completion, mental health indicators, and peer interaction patterns. A comprehensive feature engineering process is employed to transform raw educational data into meaningful learning and engagement features. These features are analysed using an ensemble of XGBoost, Long Short-Term Memory (LSTM), and Graph Neural Network (GNN) models to capture structured, temporal, and peer-related learning characteristics, thereby improving prediction reliability and robustness. To enhance transparency and support informed decision-making, the framework integrates Explainable Artificial Intelligence (XAI) techniques, including SHAP and DiCE, enabling educators to understand the factors influencing each prediction and identify suitable intervention strategies. Experimental evaluation demonstrates that the proposed approach effectively identifies students at academic risk while providing interpretable insights for teachers and administrators through timely alerts and evidence-based recommendations. The proposed framework contributes to the development of adaptive, explainable, and student-centered educational intelligence systems for future smart learning environments.

Keywords – Student performance prediction, early warning system, AI education, academic analytics, personalized learning.

I. INTRODUCTION

Artificial Intelligence (AI) has increasingly been adopted in education to improve decision-making and student outcomes. One of its most significant applications is predicting student performance and identifying learners requiring early academic support. With the growth of digital learning systems, institutions collect large volumes of data, including academic scores, attendance records, and behavioural indicators. However, these data are often underutilized for timely intervention and informed decision-making [1], [2].

Conventional assessment methods rely mainly on periodic examinations and manual observation, often identifying performance issues only after significant decline has occurred. This delays corrective actions, reduces their effectiveness, and may negatively impact student motivation and retention. Furthermore, subjective evaluation practices can introduce bias, affecting fairness in academic assessment [3], [4].

To address these limitations, recent research has focused on intelligent systems capable of analysing multi-dimensional student data for early performance prediction, enabling a shift from reactive evaluation to proactive intervention [5]. Building on this approach, the present study proposes an AI-Based Student Performance Prediction and Early Warning System that integrates academic, behavioural, and engagement-related data for continuous student risk assessment.

The proposed framework combines tree-based, sequential, and graph-based machine learning models to improve prediction reliability and capture temporal and peer-related learning patterns [5]. It also incorporates Explainable Artificial Intelligence (XAI) techniques to provide transparent and interpretable predictions for educators and stakeholders [6], [12].

In addition to prediction, the system generates automated alerts and personalized recommendations, including study plans, weak subject identification, and structured feedback mechanisms. By integrating prediction, explanation, and intervention within a unified platform, the framework supports continuous monitoring and timely academic assistance [9], [11], [14].

Overall, this research contributes toward intelligent educational systems that combine student risk prediction with actionable insights for improving academic performance and evaluation fairness.

II. PROBLEM STATEMENT

One of the major challenges in modern education is the late detection of at-risk students, especially when performance evaluation relies solely on periodic exams and informal classroom observations. In large-scale teaching environments, teachers cannot individually track every student's progress, attendance, and engagement, leading to missed opportunities for timely intervention. Moreover,

assessment bias can arise when teachers rely on qualitative impressions rather than structured, data-driven feedback, which may affect the fairness and accuracy of performance evaluation. Existing systems rarely combine academic, attendance, behavioral, and activity data into a unified AI-based framework that can automatically detect weak performers, send alerts to teachers and parents, and support personalized study and career guidance. This gap highlights the need for an intelligent, integrated system that leverages historical and real-time data to predict performance and trigger early remedial actions.

III. RESEARCH AIM

The primary aim of this research is to design and implement an **AI-Based Student Performance Prediction and Early Warning System** that analyses subject-wise marks, attendance, behavior scores, and activity participation to classify students into performance categories (Good, Average, Weak) and generate early alerts for teachers and parents. The system is designed to support schools and colleges by providing.

- A centralized student data management module for marks entry, attendance tracking, behavior rating, and activity logging.
- An AI prediction module that uses rule-based or machine learning logic to evaluate student data and assign a performance score, flagging students scoring below a predefined threshold (e.g., below 50 out of 100) as "weak" and at risk.
- An early-warning system that automatically notifies teachers and parents via simulated email or SMS, enabling timely pedagogical and home-based interventions.
- A student dashboard displaying marks, performance reports, weak subjects, and a personalized study plan, along with a career-interest module that suggests suitable career domains (e.g., Science, Commerce, Arts, Technology) based on a structured questionnaire.
- A feedback system that reduces teacher bias by using indirect and structured questions for teacher and student self-feedback, promoting fairer and more objective evaluations.

IV. LITERATURE REVIEW

A. Characteristics of Students at Risk

Students who are likely to face academic difficulties often exhibit recurring patterns over time, such as declining grades, irregular attendance, and reduced participation in learning activities. Continuous monitoring of these indicators, rather than relying only on final examination results, enables earlier identification of at-risk students [1], [3]. Previous studies have also shown that sequential data analysis methods can effectively capture temporal changes in student performance [3].

B. Factors Influencing Student Performance

Student performance depends on a combination of academic and non-academic factors. Academic records, attendance, and classroom engagement remain important indicators, while emotional well-being, social interaction, and extracurricular participation also influence learning outcomes [4]. Combining these diverse factors provides a broader understanding of student behaviour and improves prediction accuracy [2].

C. Intervention Strategies and Academic Support

Traditional educational systems often provide support only after students show poor academic performance. In contrast, recent approaches focus on early identification of learning difficulties and timely intervention. Personalized learning, targeted feedback, and continuous progress monitoring have been reported to improve student performance and engagement [7], [11], [14].

D. Challenges in Existing Systems

Although student performance prediction has advanced considerably, existing systems still have several limitations. Many approaches rely mainly on academic records or limited behavioural data, reducing the comprehensiveness of prediction models [1]. Another challenge is the limited interpretability of complex machine learning models, making prediction outcomes difficult for educators to understand and trust [4], [6]. These limitations highlight the need for more comprehensive and explainable solutions.

E. Advancements in AI-Based Prediction Systems

Machine learning methods have improved the ability to discover meaningful patterns within educational data, while ensemble techniques increase prediction reliability by combining multiple models [3], [8]. Such AI-based early warning systems allow institutions to identify students at academic risk in advance and implement timely corrective actions.

F. Role of Web-Based Educational Platforms

Web-based educational platforms provide centralized access to student information and support real-time monitoring through interactive dashboards. Role-based access enables teachers, students, parents, and administrators to interact with the system efficiently. Visualization tools further simplify complex educational data and improve decision-making [9], [13].

G. AI-Driven Identification of At-Risk Students

AI- based systems identify at-risk students by applying predictive models trained on data collected from multiple educational sources [5]. Explainable AI techniques further reveal the factors influencing predictions and suggest possible improvements. This integration of prediction and interpretability represents a significant improvement over traditional evaluation methods and supports proactive academic management [6], [11], [12].

V. RESEARCH DESIGN AND SYSTEM ARCHITECTURE

The proposed system is designed as an intelligent framework for predicting student academic risk using multi-source data, ensemble machine learning techniques, and explainable artificial intelligence. The system follows a modular layered architecture integrating data ingestion, feature engineering, predictive modelling, explainability, backend services, and role-based dashboards. As illustrated in Fig. 1, heterogeneous educational data is processed through a feature engineering pipeline and ensemble learning framework, after which predictions are interpreted using explainable AI techniques and delivered through backend APIs and alert systems.

A. Data Collection and Ingestion Layer

The system aggregates multi-dimensional student data from diverse sources to capture both academic and non-academic factors influencing performance. The primary data inputs.

- **Attendance records (daily tracking):**
- **Academic test scores (subject-wise):**
- **Behavioural indicators (discipline and engagement):**
- **Homework submission rate:**
- **Mental Health Survey (weekly 5-question form):**
- **Peer interaction graph (social network):**

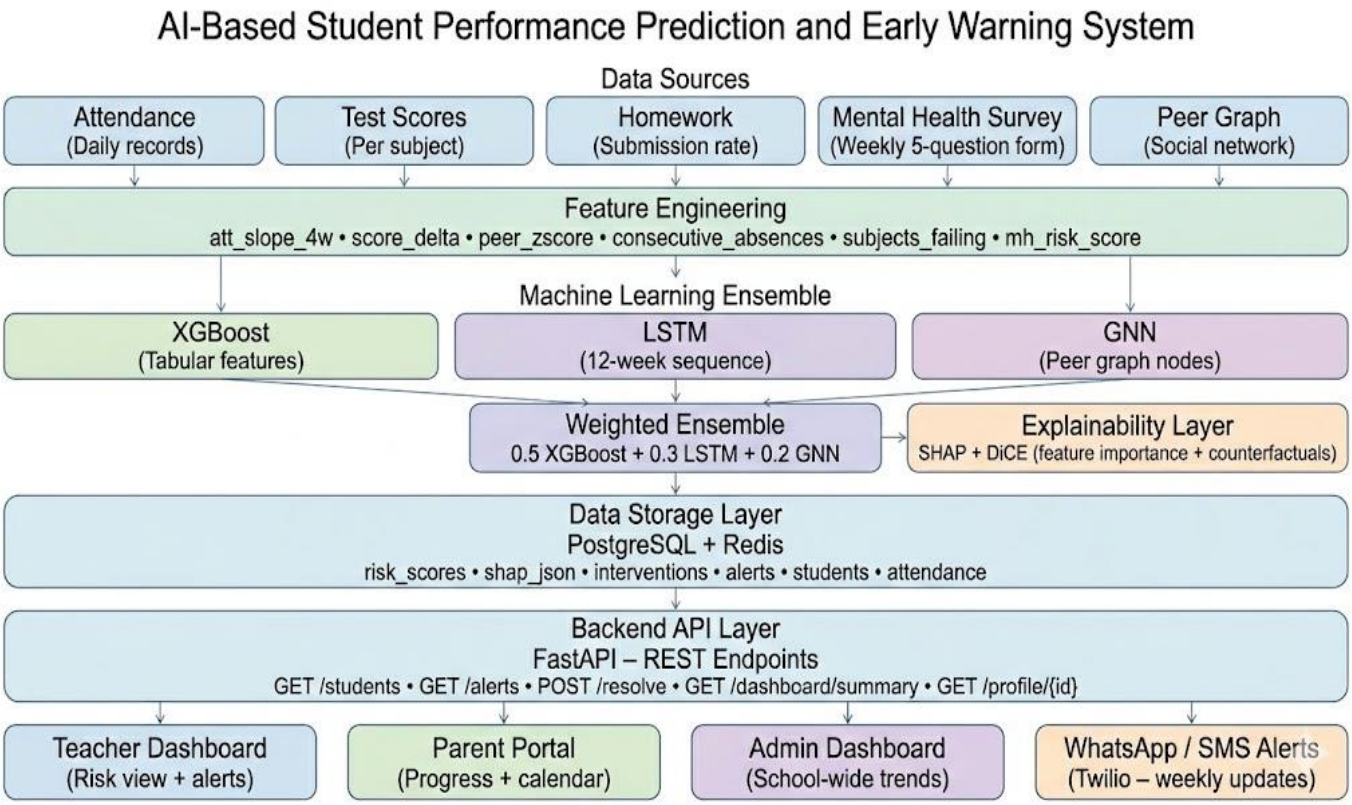


Fig. 1 – Overall System Architecture

B. Data Preprocessing and Feature Engineering

The collected data undergoes preprocessing steps including normalization, missing value handling, and temporal aggregation to ensure consistency and reliability.

To simulate realistic conditions, a synthetic dataset is generated incorporating causal relationships such as:

- decline in attendance preceding performance drop
- subject-specific sensitivity to engagement
- correlation between behaviour and academic outcomes

Feature engineering transforms raw data into meaningful predictive variables. The key engineered features include:

att_slope_4w, score_delta, peer_zscore, consecutive_absences, subjects_failing, mh_risk_score.

C. Machine Learning Ensemble Risk Engine

The core of the system is a multi-model ensemble learning framework that combines different machine learning approaches to capture diverse aspects of student behaviour.

The ensemble consists of:

- **XGBoost (Tabular Model):** Processes features such as attendance, marks, and behavioural indicators. highly effective for tabular data.
- **LSTM (Temporal Model):** Processes sequential data over a 12-week period.
- **Graph Neural Network (GNN):** Operates on peer graph data to model social relationships.

The outputs of these models are combined using a weighted ensemble approach:

$$Risk\ Score = 0.5 \times XGBoost + 0.3 \times LSTM + 0.2 \times GNN$$

Students are then classified: - High, Medium and Low Risk.

D. Explainability and Interpretation Layer

To ensure transparency and usability, the system incorporates explainable AI techniques:

- **SHAP (SHapley Additive Explanations):**.
- **DiCE (Diverse Counterfactual Explanations):**

For example, the system can indicate:

- "Low attendance and declining scores are the primary contributors to risk."
- "Improving attendance by 15% and homework completion by 10% may reduce risk level."

This layer ensures that predictions are not only accurate but also interpretable and actionable, aligning with recent advancements in explainable AI [4],[6],[12].

E. Data Storage and Management Layer

The system uses a hybrid storage architecture to efficiently manage large volumes of structured and real-time data:

- **PostgreSQL:**
- **Redis:**

The database maintains key entities such as:

student profiles, attendance records, risk scores, SHAP explanations, intervention records, alert history

This hybrid approach ensures both **data persistence and high-speed access**, which is essential for systems that continuously update predictions. Similar architectures have been widely adopted in large-scale data-driven systems [9].

F. Backend System Architecture

The backend is implemented using **FastAPI**, which provides high-performance RESTful APIs for communication between the machine learning layer and frontend dashboards.

Key endpoints include:

GET /students, GET /alerts, POST /resolve, GET /dashboard/summary, GET /profile/{id}

Additionally, FastAPI supports asynchronous request handling, enabling efficient processing of multiple prediction requests simultaneously. This is particularly important in large educational environments where real-time responsiveness is required [10].
The backend acts as the central integration layer, ensuring seamless communication between data storage, machine learning models, and user interfaces.

G. Alert and Intervention Engine

The system includes an automated alert mechanism that triggers notifications when a student is classified as high risk[13].

The system generates:

early warning alerts, recommended intervention strategies, automated notifications

Notifications are delivered via:

Email, SMS / WhatsApp (using APIs such as Twilio)

H. Application and Visualization Layer

The system provides role-based dashboards tailored to different stakeholders:

- **Teacher Dashboard:** Displays student risk levels, detailed profiles, alerts, and recommended interventions. Teachers can prioritize high-risk students and take timely action.
- **Parent Portal:** Provides simplified insights into student progress, attendance, and alerts, enabling parental involvement in academic improvement.
- **Admin Dashboard:** Offers institution-wide analytics including overall performance trends, risk distribution, and comparative insights across classes.

The dashboards are designed using modern web technologies to ensure usability, scalability, and real-time updates. Visualization plays a critical role in transforming complex model outputs into actionable insights.[13],[14] Effective dashboard design improves decision-making and user adoption in educational systems.

VI. RESULTS AND DISCUSSION

A. Experimental Setup

The proposed AI system was evaluated using a synthetic educational dataset comprising 500 student profiles and approximately 54,000 weekly records containing attendance, academic performance, homework, mental health, and peer interaction data. These multi-source records were processed through a feature engineering pipeline to derive predictive variables such as attendance trends, score variation, mental health scores, and subject failure counts

The proposed ensemble framework integrated XGBoost, LSTM, and Graph Neural Network (GNN) models to analyze structured, temporal, and peer-related data. Their outputs were combined using a weighted ensemble strategy for final risk prediction.

The dataset was divided into training and testing sets using an 80:20 split ratio. Model performance was evaluated using Accuracy, Precision, Recall, F1-score, and AUC-ROC. The proposed model achieved an AUC-ROC score of **0.813** and an average precision score of **0.752**, while cross-validation confirmed its robustness and generalization capability.

B. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to examine relationships between student behaviour, academic performance, and predicted risk levels. As shown in Fig. 2, students with higher attendance and academic scores were generally classified as low-risk, whereas lower attendance and declining performance were associated with higher academic risk.
Correlation analysis indicated that attendance, homework completion, and academic performance were negatively correlated with dropout risk, while consecutive absences and mental health stress showed positive correlation. Temporal analysis revealed a gradual decline in scores among at-risk students, supporting the use of LSTM for sequential learning. [3]. Additionally, students from lower income groups and those with higher stress levels exhibited comparatively greater academic risk.

C. Model Evaluation

The performance of the proposed AI ensemble framework was evaluated using standard classification metrics including AUC-ROC, F1-score, Average Precision, and cross-validation scores. The obtained performance metrics are summarized in Table 1.

Table. 1 – Model Performance Metrics

Metric	Value
AUC-ROC	0.813
F1-Score	0.634
Average Precision	0.752
Cross-Validation AUC	0.976
Cross-Validation F1	0.937

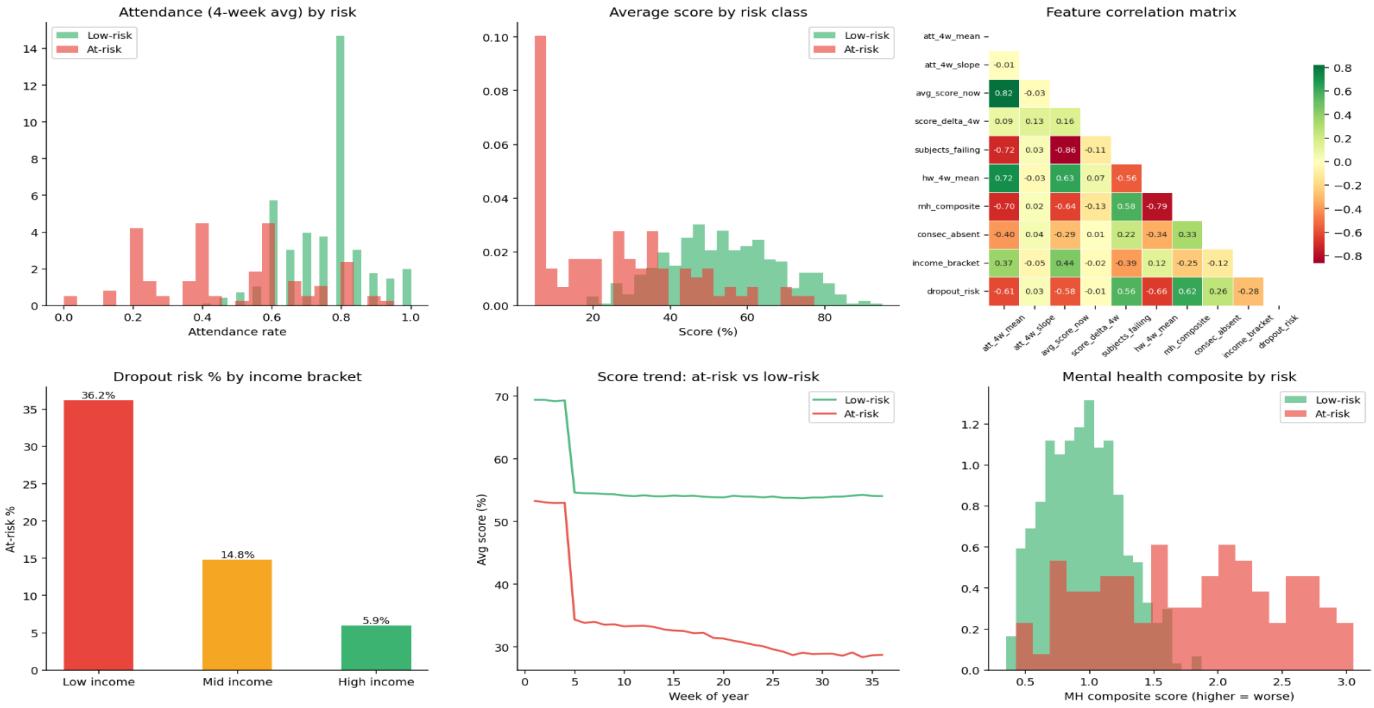


Fig. 2 – Exploratory Data Analysis of Student Risk Factors

The proposed model achieved an AUC-ROC score of **0.813** and an average precision of **0.752**, while cross-validation results (AUC = **0.976**, F1 = **0.937**) confirmed its robustness and generalization capability. The detailed evaluation results are illustrated in Fig. 3.

The ROC and Precision-Recall curves indicate that the model effectively distinguishes at-risk students while maintaining a good balance between precision and recall, making it suitable for educational early warning systems. The confusion matrix further shows that the model correctly classified **72** low-risk and **13** at-risk students, with only **8** false positives and **7** false negatives, demonstrating reliable prediction performance with relatively few misclassifications.

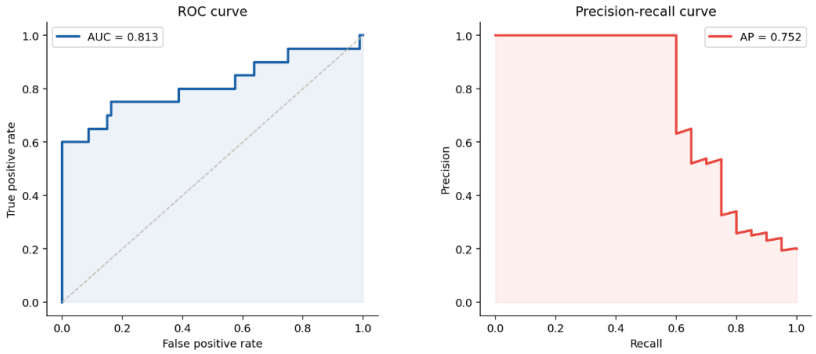


Fig. 3 – Model Performance Evaluation

D. Explainability Analysis

To enhance the transparency of the proposed prediction framework, SHAP (SHapley Additive Explanations) analysis was performed to identify the contribution of individual features toward student risk prediction [6], [12].

The global SHAP analysis revealed that academic, behavioural, psychological, and socioeconomic factors jointly influenced prediction outcomes. The most significant features included income bracket, sleep patterns, average academic score, failing subjects, and anxiety indicators, demonstrating that the proposed framework considers multiple dimensions beyond academic marks alone.

The SHAP waterfall plot further explained an individual high-risk prediction, where low sleep quality, declining academic scores, anxiety, and poor homework completion were the primary contributors to increased risk. These explainability mechanisms improve transparency by enabling teachers and administrators to better understand prediction outcomes and support informed intervention decisions.

E. Intervention Effectiveness

The proposed AI system supports intervention tracking in addition to student risk prediction. The analysis showed that counselling sessions and remedial classes were the most effective strategies for improving student performance, while teacher check-ins also contributed positively. Students receiving timely interventions demonstrated improved attendance, homework completion, and academic performance, highlighting the importance of integrating intervention mechanisms within AI-based early warning systems [7]. Furthermore, intervention tracking enables educational institutions to evaluate support strategies and make more informed, data-driven decisions.

F. Sample Student Analysis

To demonstrate the practical applicability of the proposed framework, five representative student records were selected from the generated dataset of 500 student profiles. Table II presents their academic indicators and predicted risk categories.

The sample results highlight clear differences among low-, medium-, and high-risk students. High-risk students generally exhibited poor academic performance, multiple failing subjects, and higher mental health risk indicators, whereas low-risk students maintained better attendance, stable academic performance, and fewer academic concerns. Notably, Student S003 was classified as high-risk despite good attendance, indicating that the proposed framework considers multiple academic, behavioural, and psychological

factors rather than attendance alone. These results demonstrate the practical applicability of the proposed system for providing interpretable, student-specific insights to support timely academic intervention.

Table. 2 – Sample Student Prediction Results

Student ID	Attendance (%)	Average Score	Subjects Failing	MH Composite Score	Predicted Risk
S001	80	52.4	0	1.08	Low
S002	20	21.6	4	2.58	High
S003	80	31.4	3	1.27	High
S004	60	38.7	2	0.44	Medium

G. Discussion

The experimental results demonstrate the effectiveness of integrating academic, behavioural, psychological, and temporal data for student risk prediction, indicating that academic performance cannot be accurately assessed using examination scores alone. The ensemble framework combining XGBoost, LSTM, and GNN models effectively captured structured, temporal, and peer-related learning patterns, resulting in robust and reliable prediction performance [2], [5].

The evaluation results, supported by SHAP explainability analysis, confirmed the framework's ability to accurately identify at-risk students while providing transparent and interpretable predictions for educational stakeholders. Furthermore, the integration of intervention tracking extends the framework beyond conventional prediction systems, enabling more effective and data-driven academic support [7], [11].

Overall, the proposed framework demonstrates the practical value of combining explainable AI, ensemble learning, and intervention analysis within a unified educational early warning system, supporting timely and informed decision-making for teachers, parents, and administrators.

VII. CONCLUSION

An explainable AI-based early warning system was developed to predict student academic risk using multi-dimensional educational data. The framework combines academic performance, attendance behaviour, mental health indicators, homework consistency, and peer interaction patterns to provide a more comprehensive and reliable assessment of student risk.

Experimental findings show that the ensemble framework based on XGBoost, LSTM, and GNN models can effectively identify at-risk students while maintaining strong predictive performance and model interpretability. The inclusion of SHAP-based explainability further improves transparency by helping educators understand the factors contributing to individual prediction outcomes [6].

Beyond risk prediction, the system incorporates intervention tracking and effectiveness analysis, extending its functionality beyond conventional student classification approaches. The findings suggest that integrating explainable machine learning with educational intervention analysis can enhance the practical applicability of AI-driven academic support systems.

The proposed approach can assist teachers, parents, and educational administrators in recognizing academically vulnerable students at an early stage and implementing timely intervention strategies to improve learning outcomes while reducing dropout risk.

Future enhancements may involve validating the framework using real institutional datasets, incorporating real-time classroom analytics, developing personalized recommendation mechanisms, and exploring privacy-preserving federated learning techniques to improve scalability and practical deployment in real educational environments.

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